Grade 10 maths Unit 2 Polynomial functions

Unit outcomes:

At the end of this unit, students will be able to: -

- **define polynomial functions.**
- **perform the four fundamental operations on polynomials.**
- **apply theorems on polynomial functions to solve related problems.**
- **determine the number of rational and irrational zeros of polynomials.**
- **sketch the graphs of polynomial functions.**

2.1 Introduction to polynomial functions

Competencies

At the end of this sub-unit students will be able to:

- define the polynomial functions of one variable.
- identify the degree, leading coefficient and constant terms of a given polynomial functions.
- give different forms of polynomial functions

2.2 Operations on Polynomial functions

Competencies

At the end of this sub-unit students will be able to:

- perform the four-fundamental operation on polynomials

2.3 Theorem on Polynomials

Competencies

At the end of this sub-unit students will be able to:

- state and prove the polynomial division theorem.
- apply the polynomial division theorem.
- state and prove the factor theorem.
- apply the factor theorem.
- state and prove the remainder theorem.
- apply the remainder theorem.

2.4 Zeros of a polynomial function

Competencies

At the end of this sub-unit students will be able to:

- determine the zero(s) of a given polynomial function.
- state the Location Theorem.
- apply the Location theorem to approximate the zero(s) of a given polynomial function.
- apply the rational root test to determine the zero(s) of a given polynomial function.

2.5 Graphs of polynomial Function

Competencies

At the end of this sub-unit students will be able to:

- sketch the graph of a given polynomial function.
- describe the properties of the graphs of a given polynomial function.

2.6 Applications

Competencies

At the end of this sub-unit students will be able to:

- apply polynomial function.